Mihika Swanand Sahasrabuddhe

ic24btech11014@iith.ac.in | ssmihika@gmail.com | +91 9606127141 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.01
Primus PU College	Class XII (Karnataka State board PUC)	2024	92.50%
Delhi Public School Bangalore East	Class X (CBSE)	2022	97.80%

EXPERIENCE

Summer Internship at TE connectivity, Banagalore

- Developed an understanding of end-to-end **supply chain operations** (procurement, planning, logistics, production) to contextualize forecasting outputs within broader business decisions.
- Identified limitations in the team's existing sales forecasting approach and proposed a hybrid model combining trend-based and ML techniques, improving prediction reliability for business planning.
- Structured the forecasting problem end-to-end -- from defining input variables to validating outputs against real sales data -- translating technical results into actionable insights for the team.
- Worked on a hybrid forecasting model combining **sine-wave** model with **Random Forest and XGBoost** in **Python**.
- Automated hyper parameter tuning for Random Forest and XGBoost using **Optuna**, replacing manual grid search with an **auto-incrementing optimization loop** to improve model accuracy.
- Tuned an **ARIMA** model to generate sales predictions, benchmarking performance against the hybrid ML approach.

POSITIONS OF RESPONSIBILITY

Head of design, Finance and Consulting Club IITH

- Leading a design team of about 10 coordinators along with their associates, overseeing all visual and creative output for the club. This resulted in a smoother and faster turn around of social media content.
- Part of the committee organising finance and management events across different schools and colleges in Hyderabad.
- Led the recruitment process, interviewing 20+ candidates to select a committed team for the academic year.

Coordinator in design, Finance and Consulting Club, IITH

- Managed a group of 15+ people to organise activities for the club's main event Horizon resulting in a higher student engagement.

Associate in design, Finance and Consulting Club , IITH

Head of Gesture, Fine arts and design club of IITH

- Organized workshops and events for art enthusiasts, while managing inventory and overseeing event budgets.
- Led the recruitment process by interviewing and evaluating candidates to select a committed team for the academic year.

Core member, Gesture-the fine arts and design club of IITH

- Managed club events throughout the year, increasing the participation compared to previous years.

PROJECTS

American Express Campus Challenge 2026

- Participated in Round 1 of the American Express Campus Challenge, building an **Excel-based profitability** scoring framework to **rank-order 500K** premium cardmembers using ~23 behavioural attributes (spend, revolving balance, risk, credit utilization etc.)

- Structured the problem using a revenue-vs-cost classification lens to derive a scoring formula -- applying consulting-style **segmentation** thinking to a large-scale business dataset.

Business analytics project

- Completed Finlatics' Business Analyst Program. Built a time-series forecasting model in **Excel** to project crude oil prices over a 25-year horizon using historical price data, developing **base, optimistic, and pessimistic** scenarios.
- Applied **MECE** and data-driven decision-making frameworks to real business case studies; used Power BI for analytics.

Data science projects:

- Built a house price prediction model using regression analysis and feature engineering on a multi-feature housing dataset, evaluating model performance against baseline approaches.
- Developed a movie recommendation system in Python using content-based filtering techniques on the MovieLens dataset.

SKILLS AND RELEVANT COURSES

- **Technical skills:** SQL, Python, Excel, Power BI, C++, Canva, Figma, fundamentals of AI/ML
- **Soft skills:** Analytical problem solving, structured thinking, leadership, collaboration & teamwork
- **Courses:** Introduction to probability and statistics, Machine Learning in chemistry, Introduction To Programming (C)

SCHOLASTIC ACHIEVEMENTS

- Topper in mathematics and Artificial Intelligence in class 10 CBSE boards examination.

- Won medals at school and national level in SOF Olympiads (Math, Science, and English)

EXTRACURRICULARS

- Passionate about fine arts; completed level 2 of the Pracheen Kala Kendra Fine Arts examination.
- Represented IITH in badminton for Inter-IIT 2025.
- Finished at 3rd place in a 5 km marathon in my age category.